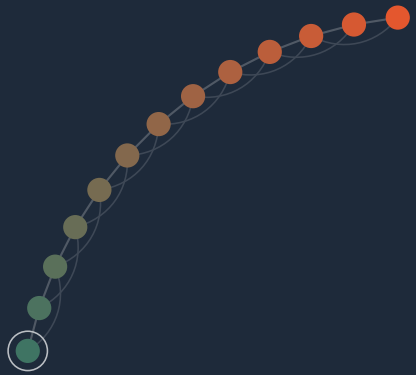
E1 277 · COURSE PROJECT · PAPER #22

Least-Squares Temporal Difference Learning

Reimplementing Boyan (ICML 1999) and a ridge-regularised extension

Member A · **Member B** · **Member C** · **Member D**

E1 277 Reinforcement Learning · November 2026



What we did

One-line summary. TD(λ) solves a linear system by slow, noisy gradient steps. LSTD(λ) estimates that system from data and **solves it directly**: same answer, less data, no step size.

Talk outline

1 Motivation & setup — Member A
2 TD(λ) and its weaknesses — Member B
3 The LSTD(λ) algorithm — Member C
4 Experiments & novelty — Member D

Course task	Our work
Understand algorithm	Derivation + paper's equivalences proved as unit tests
Run authors' code	None public (1999) $\Rightarrow$ full reimplementation, matches paper Figs. 5–6
Change settings	λ, 12 TD step sizes, feature dim. d
New environment	19-state random walk, weak features
Improvement	Ridge LSTD for very few trajectories

01

Motivation & setup

Member A

Member A

Policy evaluation

- Fixed policy π in an MDP $\Rightarrow$ a Markov chain with rewards.
- **Value function**: expected total future reward from each state

$$V^{\pi}(s) = \mathbb{E}\Big[\textstyle\sum_{t \geq 0} \gamma^t r_t \;\Big|\; s_0 = s \Big]$$

- Inner loop of policy iteration, actor–critic, STAGE, …
- If the transition matrix P is known, it is just a linear system (Bellman equation):

$$V = r + \gamma P V$$

- **In practice:** (1) P is unknown — we only see sampled trajectories; (2) too many states to store a table.

Member A

Linear function approximation

- Describe each state by K features $\boldsymbol{\phi}(s) \in \mathbb{R}^K$ and learn only K weights:

$$\hat{V}(s) = \boldsymbol{\phi}(s)^\top \boldsymbol{\theta}$$

- Covers lookup tables, state aggregation, tile coding / CMACs, RBFs, polynomials.
- **Data:** trajectories of transitions $(s_t,\ r_t,\ s_{t+1})$ until termination.
- **Goal:** find $\boldsymbol{\theta}$ such that $\hat{V} \approx V^\pi$, using as few trajectories as possible.
- **Metric (paper):** RMS error over all states

$$\mathrm{RMSE} = \sqrt{\tfrac{1}{|S|} \textstyle\sum_s \left(\boldsymbol{\phi}(s)^\top \boldsymbol{\theta} - V^\pi(s)\right)^2}$$

Member A

Monte Carlo vs. TD: the intuition



Data: 8 episodes. One starts at A and is unlucky (A→B→0). Seven start at B: six pay 1, one pays 0.

MC: target = actual return.
"From A I once got 0" $\Rightarrow \hat{V}(A) = 0$.

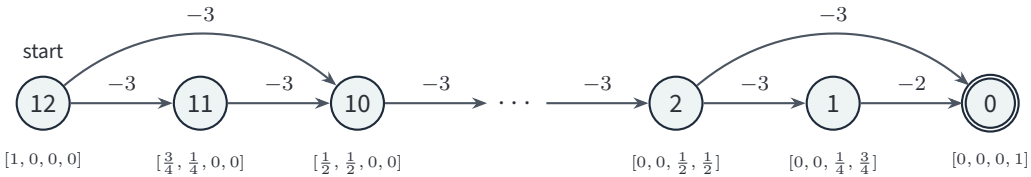
TD: target = $r + \hat{V}(\text{next})$.
"A gives 0 then B, and B averages 0.75"
$\Rightarrow \hat{V}(A) = 0.75$.

	True	MC	TD
$V(B)$	0.75	0.75	0.75
$V(A)$	0.75	**0**	**0.75**

MC: unbiased, *high variance*.
TD: *bootstraps* from its own estimate; low variance, biased while estimates are wrong.

Member A

Test domain: the Boyan chain (paper Fig. 4)



- 13 states, start at 12, state 0 terminal, $\gamma = 1$. From $s \geq 2$: go to $s-1$ or $s-2$ (prob. $\frac{1}{2}$ each), reward -3.
- $K = 4$ features: anchors at 12, 8, 4, 0; linear interpolation in between.
- True values $V^\pi(s) = -2s$ are **exactly representable**: $\boldsymbol{\theta}^* = (-24, -16, -8, 0)$.
- Every constant above checked against the paper and covered by unit tests.

02

TD(λ)

Member B

Member B

From TD(0) to TD(λ)

TD(0): nudge $\boldsymbol{\theta}$ toward a one-step target

$$\delta_t = r_t + \gamma\, \boldsymbol{\phi}(s_{t+1})^\top \boldsymbol{\theta} - \boldsymbol{\phi}(s_t)^\top \boldsymbol{\theta}, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha\, \delta_t\, \boldsymbol{\phi}(s_t)$$

TD(λ): credit each TD error to *all recent* states via an eligibility trace

$$\mathbf{z}_t = \gamma\lambda\, \mathbf{z}_{t-1} + \boldsymbol{\phi}(s_t), \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha\, \delta_t\, \mathbf{z}_t$$

- $\lambda = 0$: one-step TD (low variance, biased)
- $\lambda = 1$: Monte Carlo (unbiased, high variance)
- In between: trades bias for variance

- $\mathbf{z}_t$ propagates reward backwards *without storing the trajectory*
- Cost: $O(K)$ per step

Member B

TD(λ) in the paper (Fig. 1) and step sizes

TD(λ), offline (paper Fig. 1)

for trajectory $n = 1, 2, \ldots$
  $\Delta \leftarrow 0$, $\mathbf{z} \leftarrow \boldsymbol{\phi}(s_0)$
  for each step t:
    $\Delta \leftarrow \Delta + \mathbf{z}\left(r_t + (\boldsymbol{\phi}_{t+1} - \boldsymbol{\phi}_t)^\top \boldsymbol{\theta}\right)$
    $\mathbf{z} \leftarrow \lambda \mathbf{z} + \boldsymbol{\phi}_{t+1}$
  $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha_n \Delta$

$\boldsymbol{\phi}(\text{end}) = 0$; $\boldsymbol{\theta}$ is fixed within a trajectory.

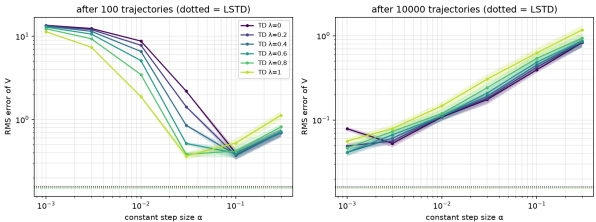
Step-size schedule (paper Eq. 8)

$$\alpha_n = a_0 \frac{n_0 + 1}{n_0 + n}$$

- $a_0 \in \{0.1,\ 0.01\}$
- $n_0 \in \{10^2,\ 10^3,\ 10^6\}$
- $\Rightarrow$ 6 schedules (paper)
- **We add** 6 constant $\alpha \in \{0.001, \ldots, 0.3\}$
- $\Rightarrow$ 12 TD configurations per λ

Member B

Weakness of TD: the step size



No single α is good both early and late ($\lambda = 0.4$, RMSE):

	$n = 100$	$n = 10^4$
TD $\alpha = 0.3$	0.70	0.85
TD $\alpha = 0.001$	13.2	0.041
TD, best sched.	0.31	0.036
LSTD	**0.15**	**0.016**

Other weaknesses (paper): sensitive to initial $\boldsymbol{\theta}$ and to feature scaling; each sample used once, then forgotten.

03

LSTD(λ)

Member C

Member C

Where does TD(λ) converge?

- Averaged over trajectories, one TD(λ) pass changes $\boldsymbol{\theta}$ by

$$\mathbb{E}[\Delta\boldsymbol{\theta}] = \alpha\,(\mathbf{d} + C\boldsymbol{\theta}), \quad \mathbf{d} = \mathbb{E}\Big[\textstyle\sum_t \mathbf{z}_t r_t\Big], \quad C = \mathbb{E}\Big[\textstyle\sum_t \mathbf{z}_t\big(\gamma\boldsymbol{\phi}_{t+1} - \boldsymbol{\phi}_t\big)^\top\Big]$$

- TD stops moving when

$$\boxed{\mathbf{d} + C\boldsymbol{\theta} = 0}$$

- C is negative definite ⇒ unique fixed point (Tsitsiklis & Van Roy).
- **TD is stochastic gradient descent on a $K \times K$ linear system** — it never writes the system down.

> **Idea:** estimate $\mathbf{d}$ and C from data and solve the system directly.

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The LSTD(λ) algorithm (paper Fig. 2)

LSTD(λ)

$A \leftarrow 0, \mathbf{b} \leftarrow 0$
for each trajectory:
  $\mathbf{z} \leftarrow 0$
  for each step t:
    $\mathbf{z} \leftarrow \gamma\lambda\mathbf{z} + \boldsymbol{\phi}_t$
    $A \leftarrow A + \mathbf{z}\,(\boldsymbol{\phi}_t - \gamma\boldsymbol{\phi}_{t+1})^\top$
    $\mathbf{b} \leftarrow \mathbf{b} + \mathbf{z}\,r_t$
whenever needed: $\boldsymbol{\theta} = A^{-1}\mathbf{b}$

- After n trajectories: $\mathbf{b} \approx n\mathbf{d}$, $A \approx -nC$
- $\Rightarrow A^{-1}\mathbf{b}$ estimates TD's fixed point
- **No step size, no initial θ**
- Invariant to feature scaling
- Keeps *all* data in A, $\mathbf{b}$
- Singular A (too little data): paper uses SVD; we use `solve` with an SVD fallback

Member C

Three views of LSTD(λ) (paper Secs. 2–3)

1. $\lambda = 0$: exactly Bradtke & Barto's LSTD (1996), with a much simpler derivation.
2. $\lambda = 1$: least-squares regression of features onto Monte-Carlo returns, built *incrementally* (telescoping: $A = \sum_t \boldsymbol{\phi}_t \boldsymbol{\phi}_t^\top$, $\mathbf{b} = \sum_t \boldsymbol{\phi}_t G_t$).
3. **Model-based view:** with one feature per state, LSTD(0) accumulates

$$A = N - C_{\text{count}}, \quad \mathbf{b} = \mathbf{s} \;\Rightarrow\; \boldsymbol{\theta} = (N - C_{\text{count}})^{-1}\mathbf{s},$$

which is exactly **"count transitions, build the empirical Markov chain, solve its Bellman equations"**.

Corridor example, one pass over the 8 episodes

LSTD(0): $\hat{V}(A) = 0.75$ (= TD's limit). LSTD(1): $\hat{V}(A) = 0$ (= MC). TD(0) needs $\sim$2000 replays with $\alpha = 0.01$ to reach 0.75.

All three equivalences are checked by unit tests in our repo.

Member C

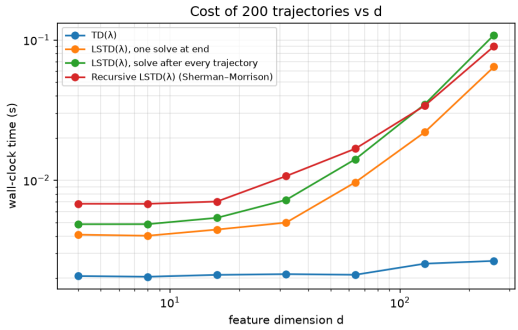
Computational cost and recursive LSTD

	per step	per solve
TD(λ)	$O(K)$	—
LSTD(λ)	$O(K^2)$	$O(K^3)$
Recursive LSTD	$O(K^2)$	—

Recursive LSTD keeps $B = A^{-1}$ up to date (Sherman–Morrison), $\mathbf{u} = \boldsymbol{\phi}_t - \gamma\boldsymbol{\phi}_{t+1}$:

$$B \leftarrow B - \frac{B\mathbf{z}\,\mathbf{u}^\top B}{1 + \mathbf{u}^\top B\mathbf{z}}$$

Start $B_0 = \delta I \Leftrightarrow$ ridge with $\epsilon = 1/\delta$.



At $d = 256$: TD 2.7 ms vs. LSTD 64 ms (200 trajectories).

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Experimental setup

Reproduction (paper Sec. 4)

- Boyan chain, $\lambda \in \{0, 0.2, \ldots, 1\}$
- Up to 10^4 trajectories
- LSTD vs. 12 TD configurations
- RMSE over all 13 states
- **20 seeds** (paper: 10), mean $\pm$ 1 SE
- Same episodes for every algorithm within a seed (common random numbers)

Implementation

- Python + NumPy, CPU only
- One module per member, frozen interfaces
- 47 unit tests, incl. paper's equivalences
- Config-driven runner, full sweep $\approx$ 30 s on a laptop

Fair comparison: TD is shown with its *best* step size — even an oracle that picks the best of 12 configurations separately at every n.

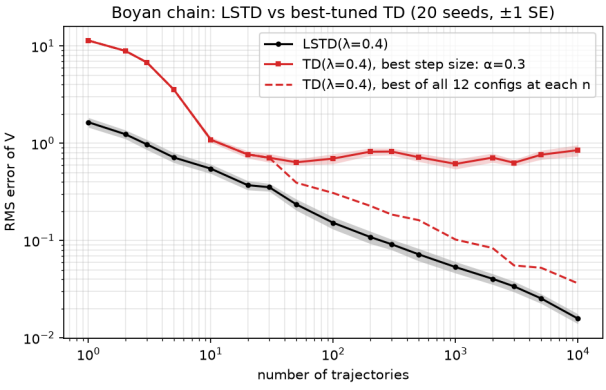
04

Experiments & novelty

Member D

Member D

Headline result: LSTD is more data-efficient



$1.64 \rightarrow 0.016$

LSTD RMSE from 1 to 10^4 trajectories

≈ 0.7

where the best single TD α (0.3) plateaus

$\approx 2\times$

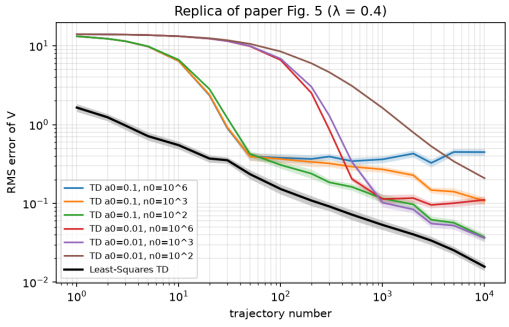
lower error than *oracle-tuned* TD (dashed) at every n

3–4×

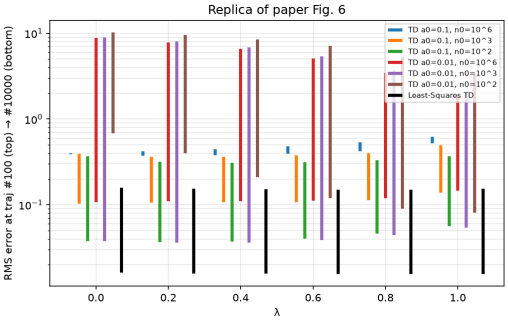
fewer trajectories for the same error

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Reproducing the paper's Figures 5 and 6



Paper Fig. 5: $\lambda = 0.4$, six schedules of Eq. 8

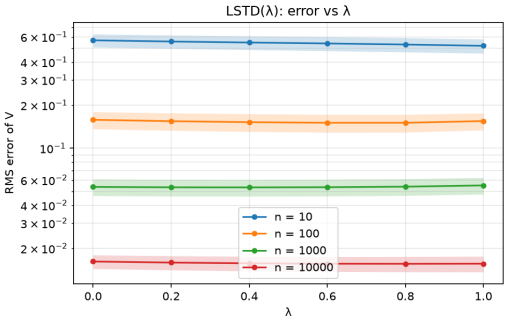


Paper Fig. 6: error at $n = 100$ (top) $\rightarrow 10^4$ (bottom)

Same conclusion as the paper: LSTD is best at every λ; TD depends critically on the schedule.

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Effect of λ



	$n = 100$		$n = 10^4$	
λ	LSTD	TD⋆	LSTD	TD⋆
0.0	**0.158**	0.368	**0.016**	0.038
0.2	**0.155**	0.318	**0.016**	0.036
0.4	**0.152**	0.308	**0.016**	0.036
0.6	**0.151**	0.312	**0.016**	0.039
0.8	**0.151**	0.329	**0.016**	0.044
1.0	**0.155**	0.366	**0.016**	0.054

TD⋆ = best of 12 TD configs for that n and λ.

- λ barely matters for LSTD here (features are exact)
- LSTD beats oracle-tuned TD by 2–$3.5\times$ everywhere

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Novelty: ridge LSTD for very few trajectories

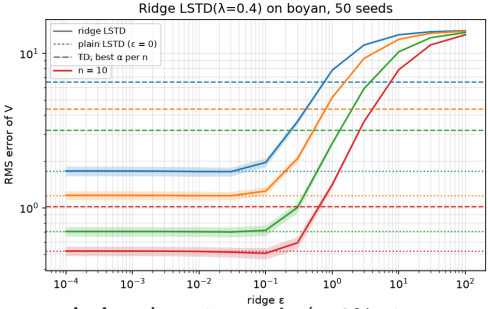
- With 1–10 trajectories, A can be singular or ill-conditioned $\Rightarrow \boldsymbol{\theta} = A^{-1}\mathbf{b}$ is unstable.
- **Ridge LSTD:**

$$(A + \epsilon I)\,\boldsymbol{\theta} = \mathbf{b}$$

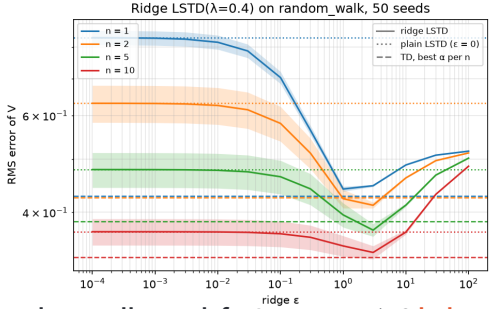
  (equivalently: recursive LSTD with $B_0 = I/\epsilon$).
- **Questions:**
  1. How does the error depend on ϵ for $n \in \{1, 2, 5, 10\}$?
  2. Can a small ϵ make LSTD competitive with the best-tuned TD at 1–5 trajectories?
  3. What happens beyond the paper: **19-state random walk** (Sutton & Barto) with **weak features** (5 aggregated groups, V^π not representable)?
- 50 seeds, $\lambda = 0.4$, $\epsilon \in [10^{-4}, 10^{2}]$; TD reference = best of 5 constant α at each n.

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Novelty: results



Boyan chain: almost no gain (1–3% at $\epsilon \approx 0.03$). A is already well conditioned; shrinking toward 0 hurts when V is down to -24. Plain LSTD already beats TD (1.7 vs. 6.5 at $n = 1$).



Random walk, weak features: $\epsilon \approx 1$–3 **halves the error at** $n = 1$ (0.83 → 0.44), matching best-tuned TD (0.43); slightly better than TD at $n = 2, 5$; TD marginally better at $n = 10$.

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Conclusions and limitations

Conclusions

- **Reproduced the paper:** LSTD beats TD at every n, λ and step size (2–$3.5\times$ below oracle-tuned TD).
- TD's best α depends on the data budget; a bad α costs 10–$100\times$.
- LSTD: no step size, insensitive to λ here.
- Price: $O(K^2)$ per step vs. $O(K)$.
- Ridge helps when data are scarce *and* features weak.

Limitations

- Toy chains; Boyan features are exact (best case for LSTD).
- TD tuned over a 12-point grid only.
- Fixed ϵ weakens as n grows (A is a sum).
- Ridge shrinks toward $\boldsymbol{\theta} = 0$, not a prior.

Next: Mountain Car with tile coding (large K) · LSPI: LSTD for control · clipped / sign TD-error baselines

Thank you

Questions?

src/ tests/ results/figures/

Backup: likely questions

Why more data-efficient? It keeps every transition in $A, \mathbf{b}$ and solves exactly. TD uses each sample for one small step and then forgets it. LSTD is effectively model-based.

What if A is singular? Too little data or redundant features. Paper: SVD pseudo-inverse. Us: `solve` with SVD fallback, or ridge $(A + \epsilon I)$.

Cost in K? $O(K^2)$ per step, $O(K^3)$ per solve (or $O(K^2)$ per step with recursive LSTD), vs. $O(K)$ for TD. Many features + cheap data $\Rightarrow$ TD can win on wall-clock time.

LSTD with $\lambda = 1$? Linear regression of features onto Monte-Carlo returns.

Control? Yes: LSTD-Q inside policy iteration = LSPI (Lagoudakis & Parr, 2003).

Is the comparison fair? Same episodes per seed; TD gets its best of 12 configs at every n.

Why does ridge not help on Boyan? A is well conditioned, and shrinkage biases $\boldsymbol{\theta}$ toward 0 while $\boldsymbol{\theta}^* = (-24, -16, -8, 0)$.

Backup: why LSTD(1) = Monte-Carlo regression

With $\lambda = \gamma = 1$, $\mathbf{z}_t = \sum_{i \le t} \boldsymbol{\phi}_i$ and $\boldsymbol{\phi}_T = 0$ at termination:

$$A = \sum_t \mathbf{z}_t (\boldsymbol{\phi}_t - \boldsymbol{\phi}_{t+1})^\top = \sum_i \boldsymbol{\phi}_i \sum_{t \ge i} (\boldsymbol{\phi}_t - \boldsymbol{\phi}_{t+1})^\top = \sum_i \boldsymbol{\phi}_i \boldsymbol{\phi}_i^\top$$

$$\mathbf{b} = \sum_t \mathbf{z}_t r_t = \sum_i \boldsymbol{\phi}_i \sum_{t \ge i} r_t = \sum_i \boldsymbol{\phi}_i G_i$$

So $\boldsymbol{\theta} = A^{-1}\mathbf{b}$ is ordinary least squares of G_i on $\boldsymbol{\phi}_i$ — built incrementally without storing trajectories.

Backup: code $\leftrightarrow$ paper

Paper	Code	Owner
Fig. 4 chain and features	`env.BoyanChain`, `features.boyan_features`	A
Exact V^{π}	`env.true_values` (Bellman solve)	A
Fig. 1 TD(λ), Eq. 8	`td_lambda.td_lambda_curve`, `step_size`	B
Fig. 2 LSTD(λ)	`lstd.lstd_lambda_curve`, `solve_theta`	C
Recursive LSTD (Sec. 2)	`lstd.recursive_lstd`	C
Sec. 3 model-based view	`tests/test_lstd.py`	C
Figs. 5, 6 and our figures	`run_experiment.py`, `plot_results.py`	D

```
python -m src.run_experiment && python -m src.plot_results
```